

Kit Unboxing & First Electronics

How to teach this class — what happens, how to run it, and why it matters. The day the kit becomes a robot.

THE ONE GOAL

The Triggerfish comes out of the box, gets counted, and takes its first real connections — and the season's records open with their first entries.

HOW THIS CLASS IS TAUGHT — YOU FACILITATE, THE KIDS BUILD

The concepts were taught in Class 1.2 (soldering) and stay pre-watched. This class is almost entirely hands-on, run from the **Triggerfish build manual** — the kids inventory, read, and wire while you keep the room safe and the records honest. One three-hour session.

BEFORE YOU TEACH THIS

- **Read the manual's parts list and build sequence** so inventory and the first stage move fast.
- **Have the soldering stations ready** — irons get hot again for the first kit connections.
- **Set up the three records** — know how to make a GitHub org, and have the notebook and a camera ready.
- **Read the capsule call-out below** so you can explain what the kids are designing and why.
- **Confirm the battery is on order** — the kit ships without a power source, needed before the wiring lessons.
- **Section 4.3:** the kids count, read, wire, and record — you keep it safe and honest. Hands off.

WHY THIS CLASS MATTERS

This is the day the kit stops being a box of parts and starts being a robot. Inventory discipline now — counting every part against the manual — is what keeps a missing screw from costing a week mid-build. The first real connections put the soldering skill straight onto the vehicle. And the three records that open today — the repository, the notebook, and the photo log — are the quiet documents that win points in April, but only if they start now and run honestly.

HOW THE MEETING RUNS

0:00 · 15 MIN Welcome & safety brief

Open with the standing brief — today, hot irons again plus small parts and sharp tools. Name them: irons in their stands, fume fan on, glasses down; count and bag small parts so none walk off; wash hands after solder. The brief is short, but the room has real hazards in it.

0:15 · 35 MIN Unbox & inventory the kit

Hand the manual to the Mechanical Lead and let them run inventory against the parts list — don't do it for them. The one rule: count everything before building. Flag it out loud — **the kit ships without a power source**, so confirm the battery is on order. A missing part found today is an email; found mid-build, it's a lost week.

0:50 · 20 MIN Read the build sequence

Walk the manual's build order together, start to finish, before touching a part. Group the parts for the first stage and match each to the diagram. Ask questions now, not mid-solder — knowing the whole path keeps the build calm.

1:10 · 50 MIN First real connections

Irons are hot again. The joints the kids practiced in 1.2 become real connections on the robot now — stage by stage, every member soldering a real section. Stop at a clean point when the clock runs out; never rush a connection. Photograph each stage for the log as you go.

2:10 · 25 MIN Open the three records

Stand up the three records that win points in April: the Software Lead creates the GitHub org on student accounts, the team writes one real decision with its reason in the notebook, and the photo log opens with today's pictures. Assign an owner to each, out loud. A notebook written as you go is honest; one written from memory is fiction — and judges can tell.

2:35 · 15 MIN Design the capsule

Each kid sketches a small waterproof capsule — a sealed PVC body, one cable through the wall, and a plan to keep a paper towel inside dry. Design only today: pencil and paper, no building. Let them propose; don't engineer it for them. Sketches are due before they leave.

WHAT THE CAPSULE ACTUALLY IS — READ THIS SO YOU CAN EXPLAIN IT

The capsule is a small **sealed PVC tube** with **one cable passing through the wall** — a “penetration.” That single hole is exactly where water wants to get in, and sealing it is the hard part of waterproofing anything underwater. **The paper towel is the test instrument:** seal one inside, dunk the capsule, then open it — dry means the seal held, damp means it leaked and they trace it to a cause. It matters because it is the **foundation skill for the Phase 2 vertical profiling float**, which must stay watertight at 2.5 meters. Designed today; cut, sealed, and dunked in later lessons; the Phase 1 gate is three consecutive dry dives.

2:50 · 10 MIN Wrap-up & homework

Close the loop: finish the capsule sketch, confirm GitHub access, and watch the Class 1.4 pre-watch (Multimeter Basics) when it's assigned. On the adult track, make sure the battery and the SeaMATE Simple Circuits Lab Kits are on order before the electronics workshop, and keep the pool prerequisites moving.

THE PARENT LANE — SECTION 4.3

The kids count the parts, read the manual, solder the connections, and write the records — all of it. You keep the room safe and the records honest, and you photograph the build. The robot the kids build, redos and all, is the robot that races. Disclose any outside help, including AI, in the documentation.

COMMON PROBLEMS & QUICK FIXES

SYMPTOM	WHAT'S WRONG & THE FIX
A part seems missing	Re-check the bag and the manual's count before declaring it lost. If it's truly missing, log it and email SeaMATE the same day — don't build around the gap.
Solder won't take on a kit pad	Same as 1.2: tip not tinned or the pad not hot/clean. Re-tin, heat the pad first, then feed solder.
Build is getting chaotic	Stop and re-read the stage in the manual together. One stage at a time, matched to the diagram, beats racing ahead.
Notebook entry is vague	Push for the <i>why</i> , not just the <i>what</i> : "we chose X because Y." That's what scores in April.

WHAT "DONE" LOOKS LIKE

- The Triggerfish inventoried against the manual, every part accounted for.
- The first kit connections soldered — electronics underway.
- Repository, notebook, and photo log all open with their first entries.
- A capsule design on paper from every kid.

ONE SESSION — PROTECT THE INVENTORY

If the clock gets tight, never skip the inventory or rush a connection — those are the failures that cost a week later. Let the records or the capsule sketches carry a little into homework instead. The kit comes alive next in the electronics workshop, Class 1.4.